Experiment setup for a deterministic Bunch-to-Bucket transfer for FAIR

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Abstract

The FAIR accelerator facility is expected to be one of the largest ion research centers worldwide. In the frame of this project, the synchrotron SIS18 will serve as an injector for the SIS100, currently under construction. The development of a deterministic Bunch-to-Bucket synchronization procedure is one of the key features of the beam transfer system between SIS18 and SIS100.
The Bunch-to-Bucket transfer for FAIR must be scalable to any type of ion, which are accelerated in the SIS18. Two extreme cycles are taken into account in the frame of this research: proton beam up to an energy of 4 GeV and uranium ions U^{28+} up to an energy of 196 MeV/u at transfer flattop in the SIS18.
An experimental setup aiming at validating the low level radio frequency (LLRF) topology for the phase advance acquisition and the phase advance control procedures has been planned.

Introduction

In the framework of the FAIR complex, hadrons from protons to uranium will be accelerated by a chain of accelerators composed mainly of the UNILAC, the SIS18 and the SIS100. A series of further synchrotrons will extend the initial chain: the SIS300, the HESR, the CR and the NESR.

One of the key features of the FAIR project is the beam transfer technology, under development, the Bunch-to-Bucket transfer, which consists in extracting the beam from the source synchrotron within one revolution and capturing it at the injection within the target synchrotron.

In order to perform a Bunch-to-Bucket transfer, a preliminary synchronization of the source and the target synchrotron is required. The synchronization of two Synchrotron RF systems results in controlling the phase advance relation between their driving RF signals [1].

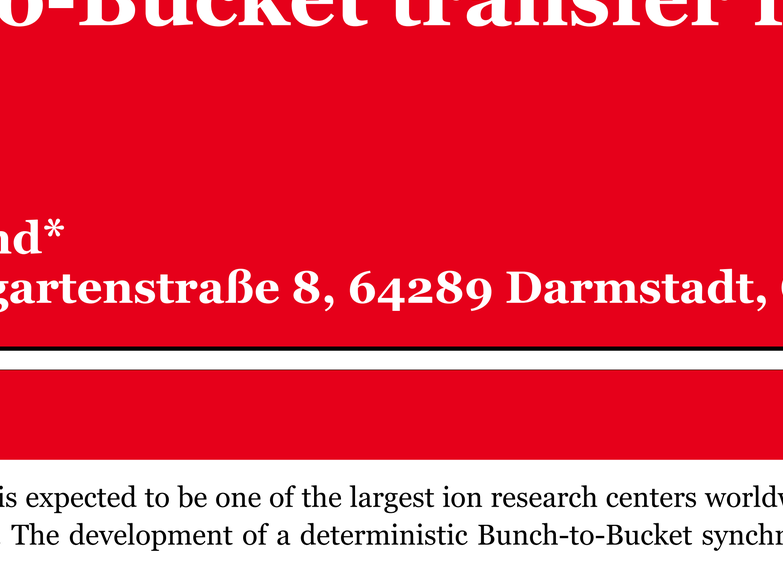


Figure 1: International Facility for Antiproton and Ion Research (FAIR) at GSI, Darmstadt.

A Multi-purpose Hardware Unit

The LLRF topology for FAIR is based on local signals generation by means of the DDS principle. The BuTiS time base enables to synchronize the signal generation. Feedback and control loops are implemented by means of a DSP phase discriminator.
To perform the Bunch-to-Bucket synchronization, three additional LLRF functions are required:

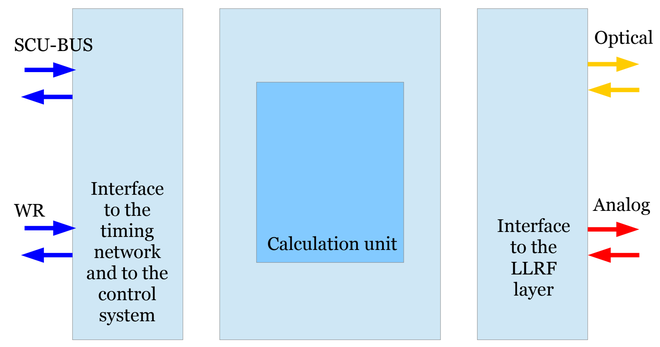


Figure 2: Principle of the MHU.

- Phase Advance Prediction (PAP).
  The DSP-based phase measurement is performed asynchronously from the BuTiS time base for technological reasons [2]. The phase advance prediction function aims at determining a reference phase advance value at a precise BuTiS time and at forwarding this value to the Timing Network, which ensures the data transfer between the distributed actors of the synchronization, through a Scalable Control Unit (SCU) [3].
- Phase Correction Module (PCM).
  This function aims at resynchronizing distributed copies of a reference RF signal, for instance a copy of the target h=1 signal in the supply room of the source synchrotron.
- Phase Shift Module (PSM).
  This function aims at modulating a DDS output RF signal adiabatically according to its current frequency and to the phase advance value to compensate.

To perform these specific functions, a new Multi-purpose Hardware Unit is being developed at GSI in cooperation with the IES institute (TU Darmstadt). This MHU fulfills the role of interfacing the timing network (White Rabbit, SCU-bus) with the LLRF layer (see Fig.2).

Topology of experiment

The experiment setup is expected to emulate the overall synchronization procedure between two synchrotrons. In the framework of a Bunch-to-Bucket transfer, the receiver synchrotron is set to its injection energy. The beam is accelerated in the injector until it reaches this energy (see Fig. 3). Once the acceleration is complete, the relative phase advance between the reference RF signals of both synchrotrons is evaluated and corrected before the beam is kicked out.

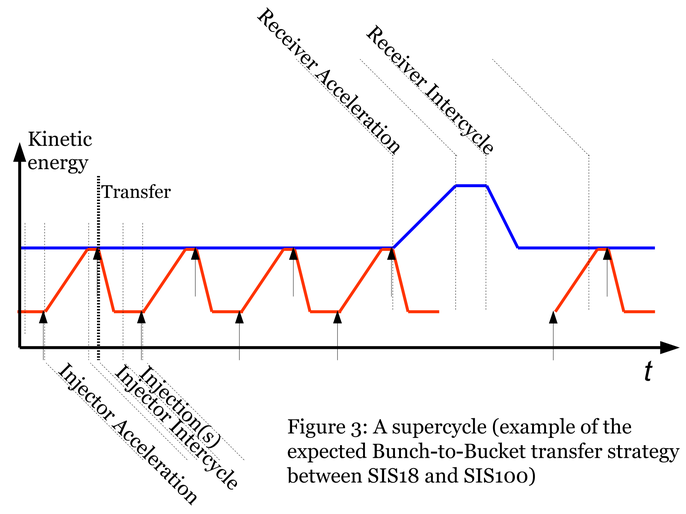


Figure 3: A supercycle (example of the expected Bunch-to-Bucket transfer strategy between SIS18 and SIS100)

The reference signal generation and the phase advance measurement are performed locally in the supply room of the corresponding cavity units. Between two supply rooms, the data are transferred through the Timing Network by means of the White Rabbit system (see Fig. 4 [4-5]). The further functions described above (the phase advance prediction, the phase shift and the phase correction functions) are thus performed locally, close to the LLRF module, with which they are communicating.

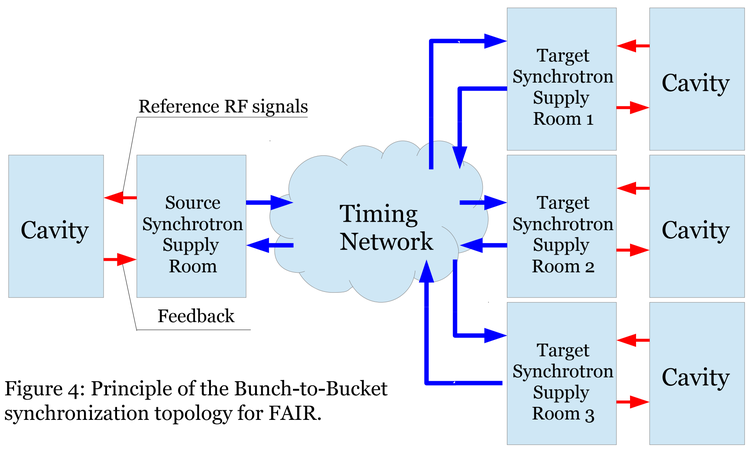


Figure 4: Principle of the Bunch-to-Bucket synchronization topology for FAIR.

In the framework of the synchronization topology, the phase advance between the group DDS signal of corresponding harmonic number of the source and the target synchrotron must be measured relatively to the same reference RF signal by means of a DSP system. A phase advance prediction module is implemented for each of the two phase advance measurement systems.
For validation purposes, the phase correction and the phase shift modules must be implemented only once, locally near the LLRF modules, which they must control.
The communication between the two emulated supply rooms is supported by a delayed optical fiber.
The overall experiment control is ensured by means of a computer (see Fig. 5).

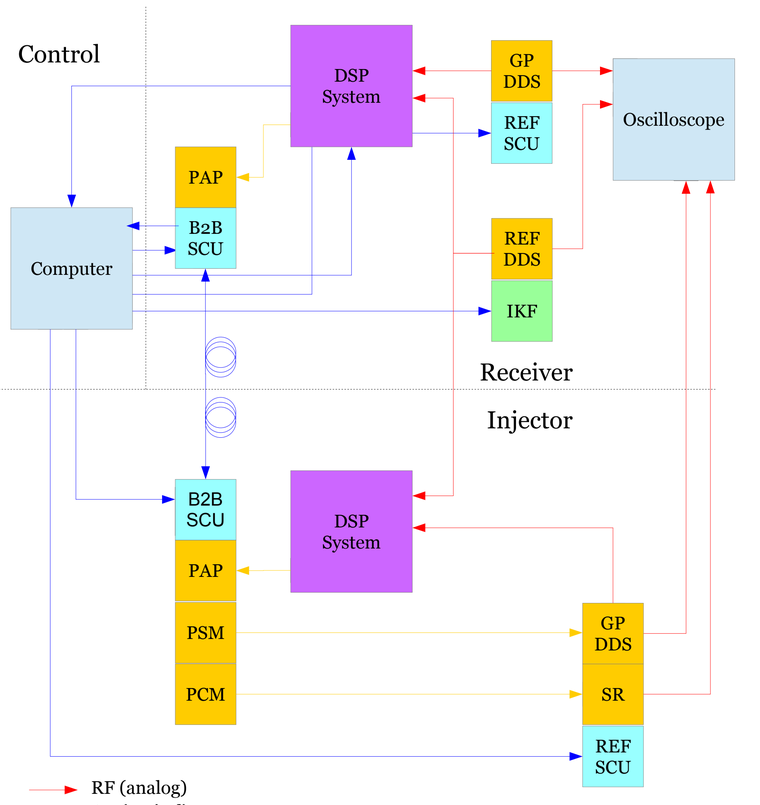


Figure 5: Proposal of an experiment setup for the Bunch-to- Bucket synchronization topology of FAIR.

References:
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